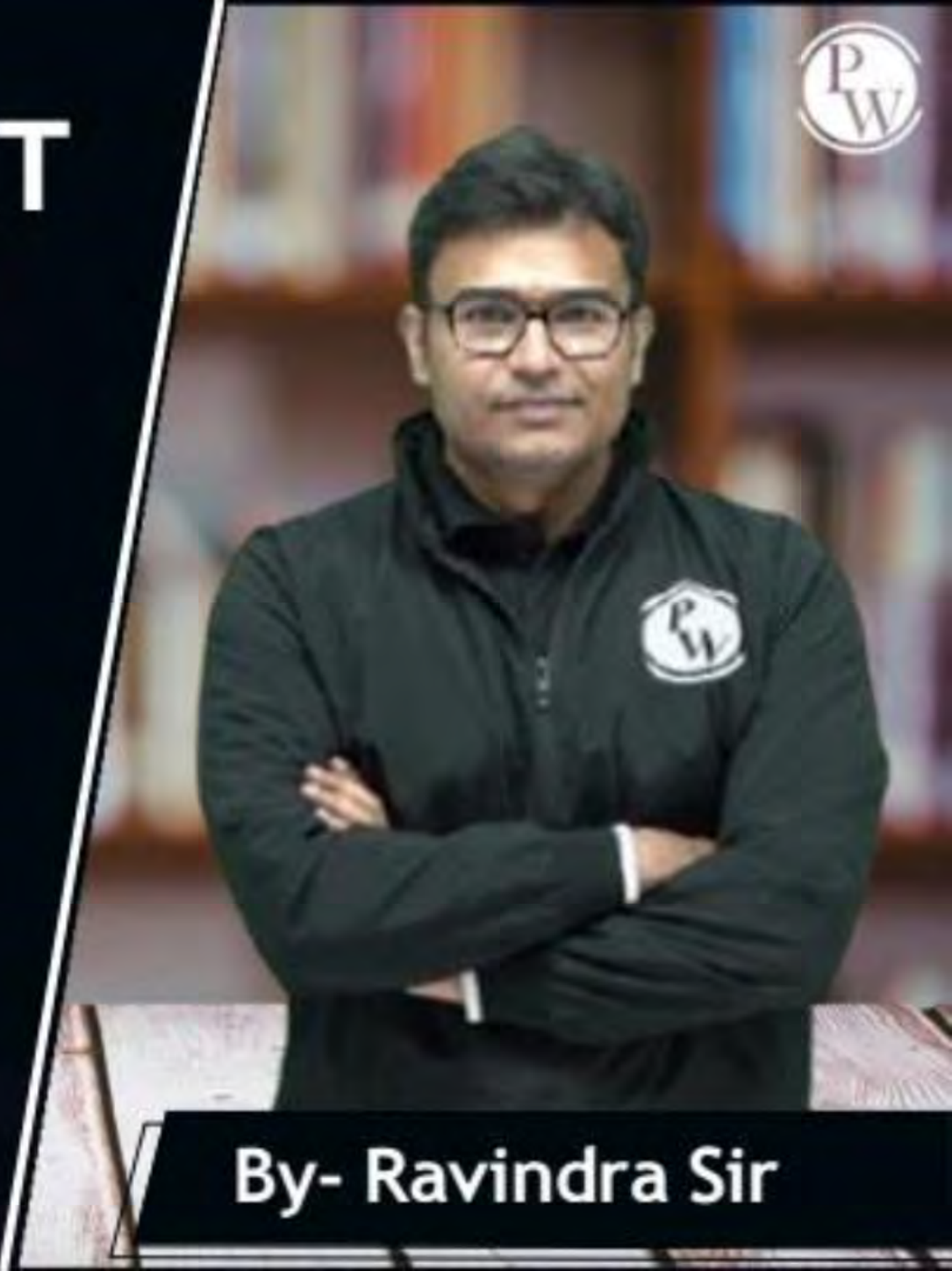
Computer Science & IT

COMPUTER NETWORK
(CN)

Hardware and Various Devices Used in Networking

Lecture No. 01



By- Ravindra Sir

Recap of Previous Lecture



Topic

Topic

Topic

Firewall

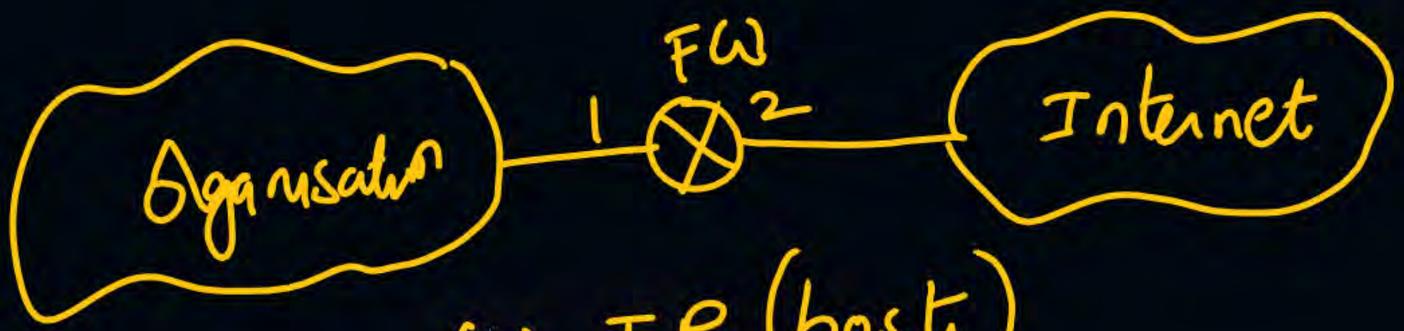
Topics to be Covered



- Topic
- Topic
- Topic

DNS

Layer 5 FW & proxy FW



(i) IP (hosts)

(ii) Port (services)

(iii) Service on a host

(iv) AL (username, Password)



Rule table :-

SIP	DIP	SP	DP	Interface	User
ICS	—	21 → FTP	—	1	—
—	ICS	—	23 → Telnet	2	Non-employees
—	—	80 ✓ → HTTP	—	2	—

DNS (Domain name System)

Humans are comfortable with hostnames.

Host names does not have routing information

So we a directory service to convert hostname → IP

This is done by DNS

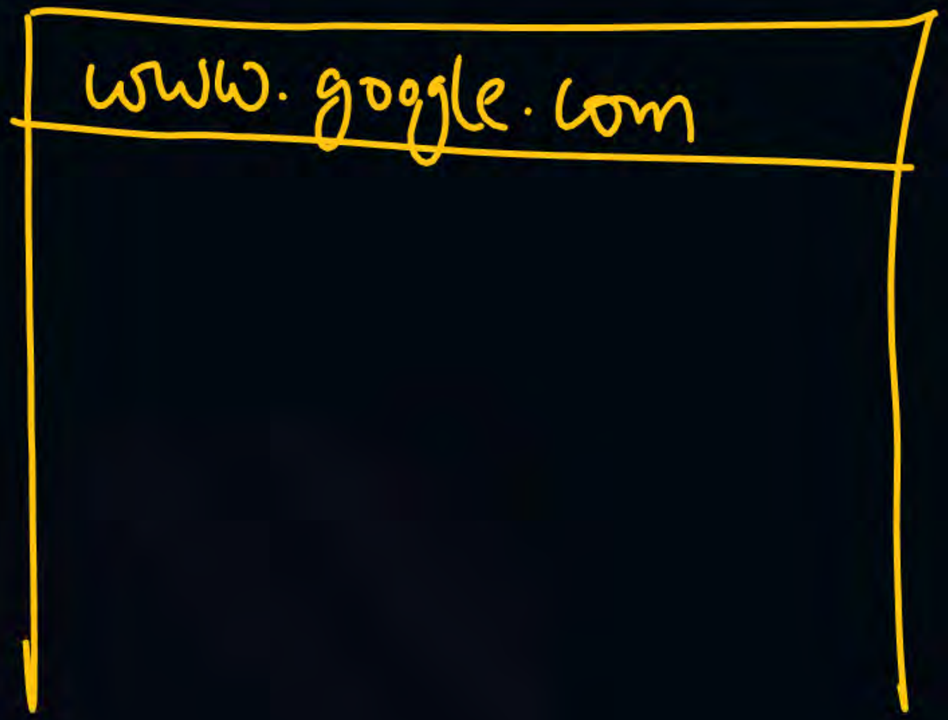
The main task of Internet's DNS is:-

① A distributed database implemented in a hierarchy of DNS servers

② An application layer protocol that allows hosts to query the distributed database.

→ DNS uses UDP and port number 53

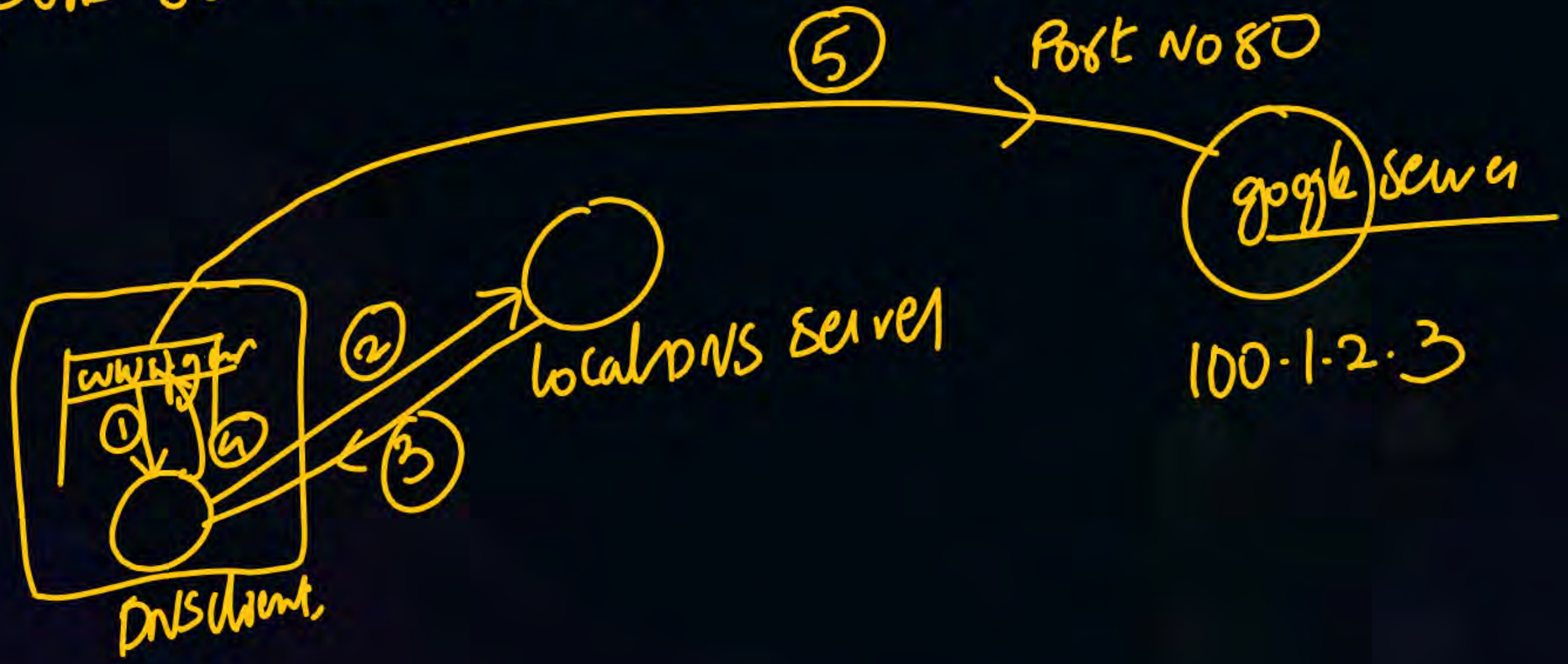
DNS is Commonly employed by other application layer protocols like HTTP and SMTP to translate user supplied host names to IP addresses


www.google.com

Assume a host needs IP address of www.google.com

1) The same user machine runs a client side of DNS application
2) The browser extracts the host name www.google.com from the URL and passes the host name to client side of DNS application
3) The DNS client side sends a query containing the hostname to the DNS server
4) The client side eventually receives a reply with IP address for www.google.com

5) once the browser receives the IP address from DNS client, it can initiate a TCP connection to the HTTP server at port 80 at that IP address



DNS provides few important services in addition to translating host → IP

(1) Host aliasing :- A host can have a complicated host name with one of more alias name

Ex :- relay1.westcost.google.com → Canonical name

alias – www.google.com

DNS can be invoked by an application to obtain canonical host name for a supplied alias name as well as the IP address of the host

② mail Server aliasing :-

RBR@yahoo.com ✓

alias ← yahoo.com ✓ ws serve : relay1.westcoast.yahoo.com

mail server Canonical name → mail.yahoo.com

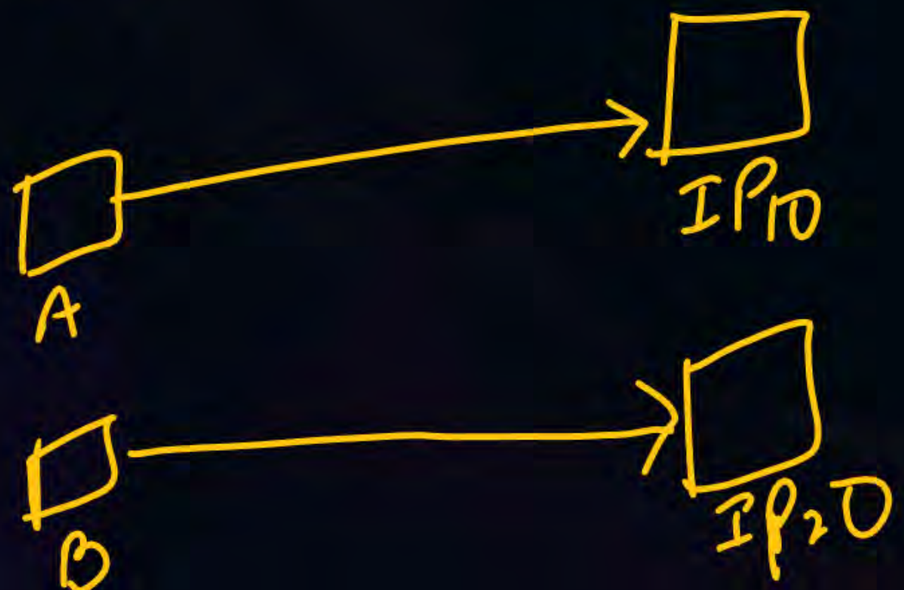
DNS can be involked by the mail application to obtain Canonical name of the Supplied alias name. as well as IP address of the mail server. In fact (MX) record permits a Company's mail Server and webserver to have the same alias name ✓

③ local balancing :-

google.com → IP_1 IP_2 IP_3 IP_4 - - - - IP_n

A → google.com ← IP_{10} IP_7 I_8 IP_{20} - - IP_n

B → google.com ← I_{20} I_1 I_{10} I_9 - - - - IP_n



DNS is also used for load distribution among replicated servers, such as replicated web servers
Busy sites like google.com are replicated over multiple servers with each server running a different machine and having a different IP address. For replicate web servers, a set of IP addresses are associated with one alias host name. The DNS database contains this set of IP addresses, the DNS server responds with the entire set of IP addresses but rotates the ordering of the addresses in the DNS Reply.

Because a client typically sends its HTTP request to the IP address that is listed first in the set, DNS rotation distributes the traffic among the replicated servers. The DNS rotation is also used for email servers so that multiple mail servers can use the same alias name

How DNS works :-

Disadvantage of central DNS server :-

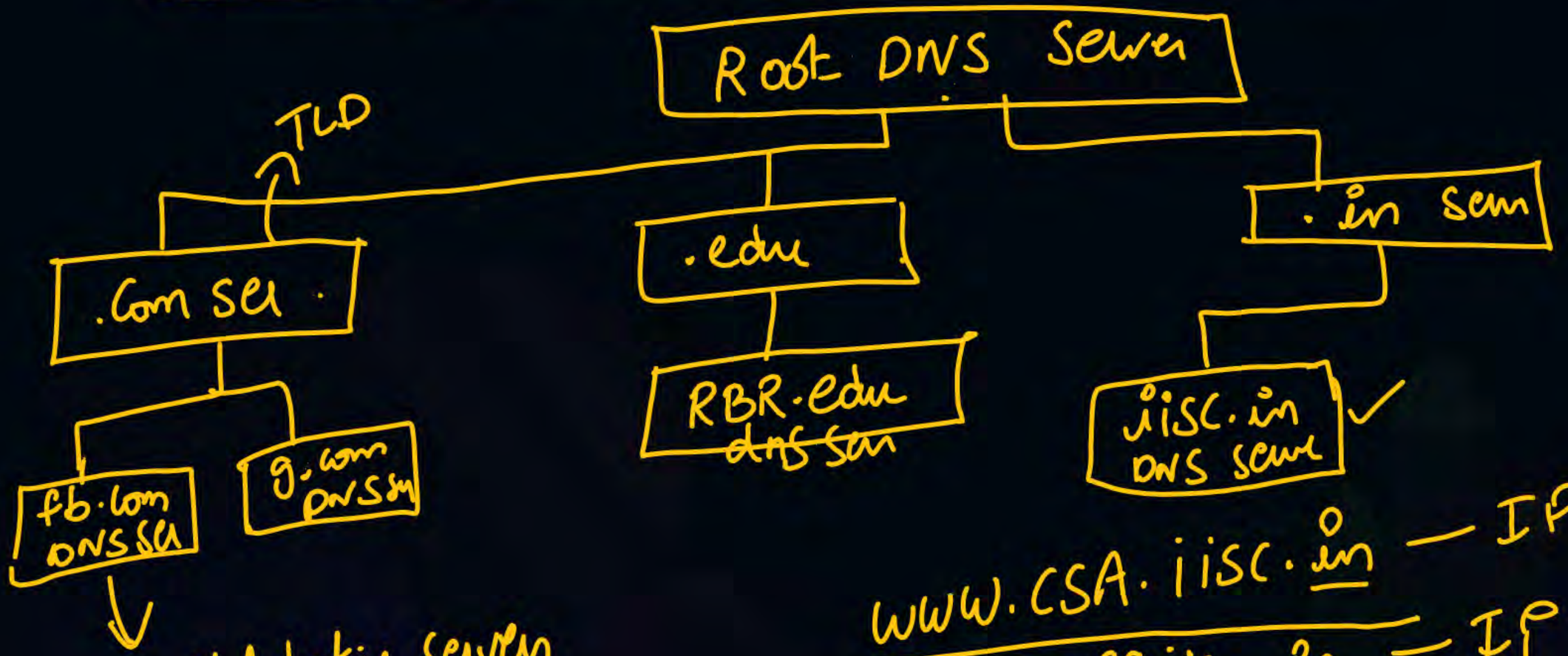
→ single point failure

→ Traffic Volume

→ Distant centralized DB

→ maintenance.

A distributed DNS Servers, hierachial DB



www.CSA.iisc.in — IP

www.SERC.iisc.in — IP

The DNS servers are often unix machines running the S/W The berkeley Internet name domain (BIND)

In order to deal with the issue of scale, the DNS uses a large number of servers, organised in hierarchial fashion
No single DNS server has all the mappings of all the hosts in the world. Instead the mappings are distributed across the DNS servers. There are 3 classes of DNS servers

1) Root DNS servers
2) Top level Domain (TLD) servers
3) Authoritative DNS servers

Root DNS Server:-

There are more than 1000 root server instances scattered all over the world. These are manged by 12 different organisations and coordinated through Internet assigned name authority (IANA). Root Servers provide the IP addresses of TLD servers

Top level domain (TLD) Servers :-

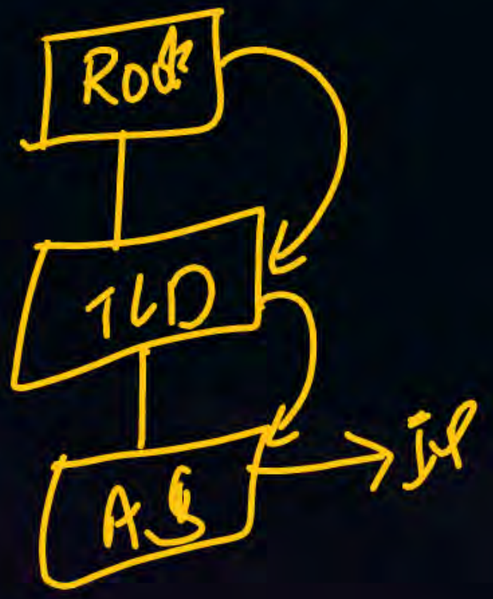
For each top level domain such as com, org, net, edu and all country level domains like in, UK, fr, US there is a TLD server

The company Verisign global registry Services maintains TLD Servers for com top level domain and company Educause maintain TLD for edu top level dom

The TLD Servers provide IP addresses for authoritative DNS Servers

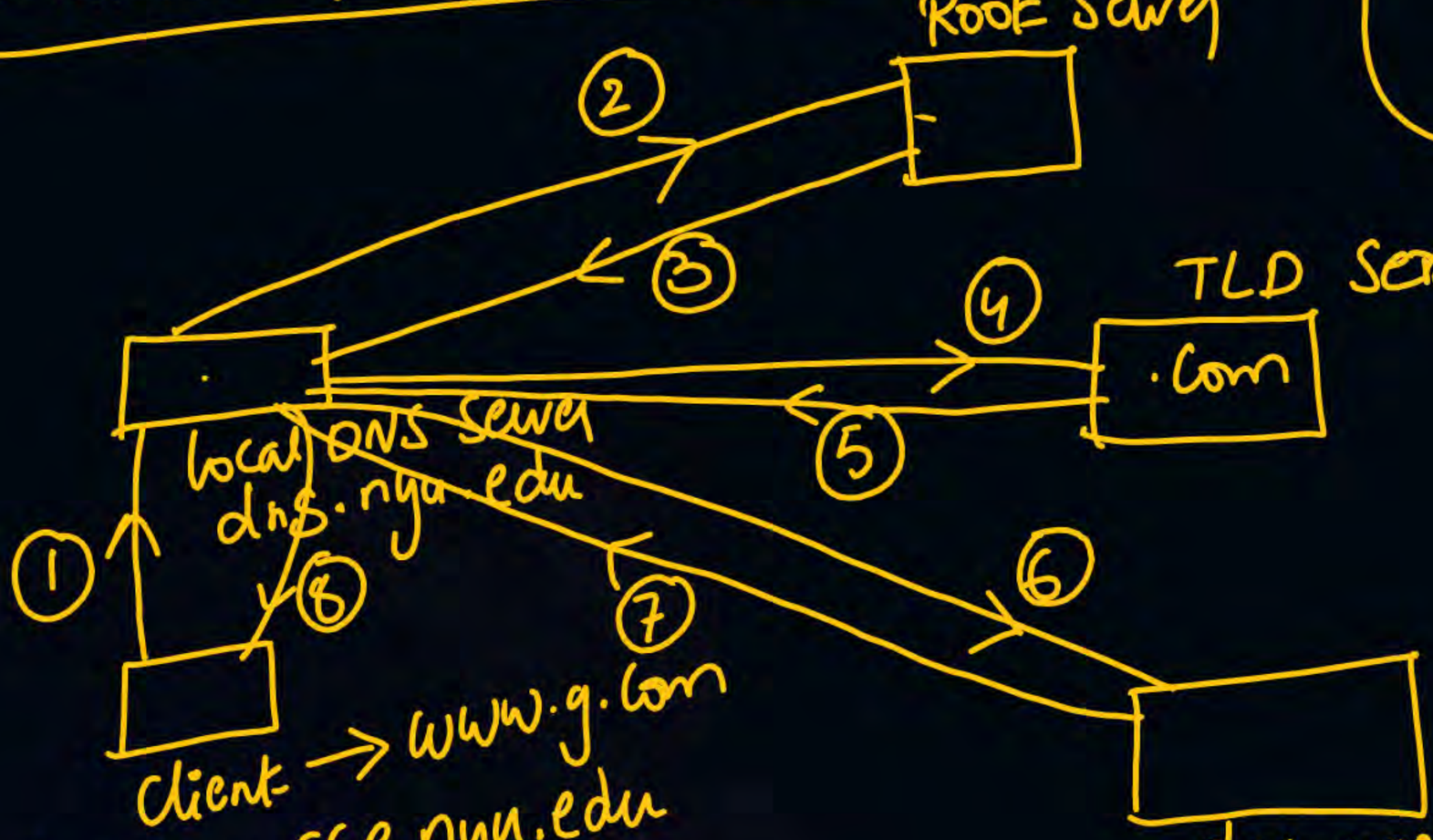
Authoritative DNS Servers :-

Every organisation with publicly accessible hosts (Such as webservers and mail servers) on the internet must provide publicly accessible addresses. The organisation's authoritative DNS server houses these DNS records



→ There is another important type of DNS called local DNS server. This server doesn't strictly belong to the hierarchy of servers but it is central to the DNS organisation. Each ISP provides local DNS server through DHCP. A local DNS server is typically close to the host. It may be on the same LAN.

Interaction of various DNS servers :-



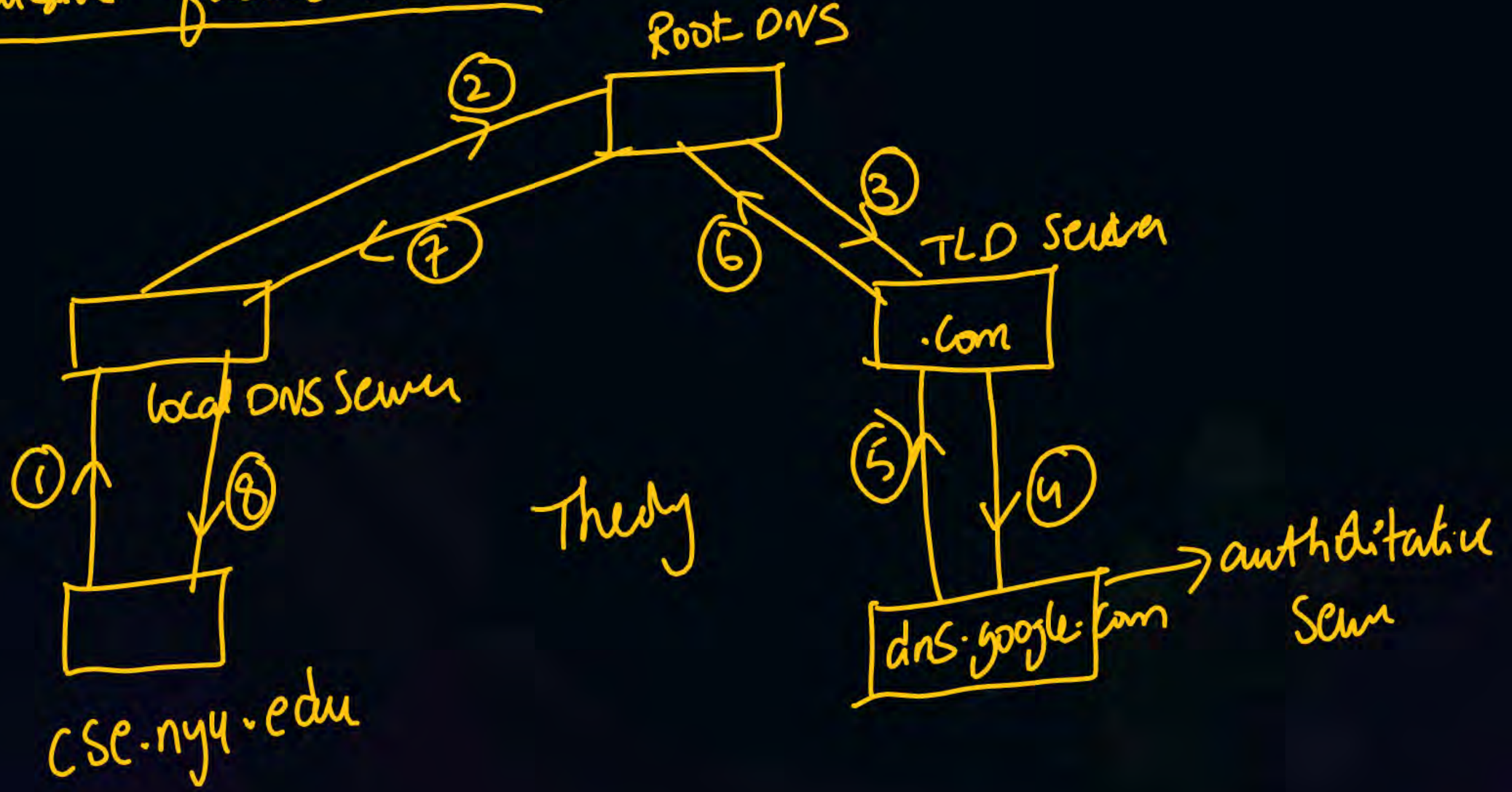
Practically

It is a combination of both recursive and iterative queries

First request is recursive and later 3 queries are iterative

In theory any DNS query can be iterative or recursive

Recursive queries in DNS



Theory

DNS Caching :-

DNS extensively exploits DNS caching to reduce the no of DNS queries

For ex :- Each time a local DNS server dns.nyu.edu receives a reply from some DNS server, it can cache any information contained in the DNS reply.

Of hostname/IP address pair is cached in a DNS server and another query arrives to the DNS server for the same hostname the DNS server can provide the IP address even if it is not authoritative

Because IP addresses are not permanent, DNS servers discards the cached information after a period of time (after 2 days)

A local DNS server can also cache the IP address of TLD servers thereby bypassing root DNS. So root servers are bypassed, for all but a small fraction of DNS queries.

THANK - YOU